

API Reference

Generated from docs/api-reference.md

This document is the API reference for the agent marketplace backend.

The implementation is HTTP-first, versioned under /v1 for domain routes, and uses FastAPI-style JSON bodies and status codes.

Common Mechanics

Base URLs

- Health routes: /health, /health/live, /health/ready
- Domain routes: /v1/...

Common Headers

- Authorization: Bearer <jwt-or-api-key> for protected routes that accept generic bearer auth
- Idempotency-Key for invoke requests and provider payout requests
- X-Request-ID for request correlation
- PAYMENT-REQUIRED on 402 Payment Required responses
- PAYMENT-SIGNATURE on paid retry requests produced by the x402 client
- PAYMENT-RESPONSE on successful paid invoke responses
- The service uses X-Request-ID for request correlation and echoes it when present

Common Error Shape

Most route handlers raise FastAPI HTTP errors with JSON bodies shaped like:

```
{  
  "detail": "human-readable message"  
}
```

Authentication Matrix

- Public: health routes, discovery routes, quote creation, and auth nonce / verify / refresh
- Generic bearer auth: provider authoring, invoke, invocation reads, provider earnings, ledger, provider payout reads, and admin routes
- JWT-only: API-key lifecycle, account self-service, wallet rotation, and provider payout execution

Health

GET /health

- Auth: none
- Request: no body, no parameters

- Success: 200 with {"status": "ok"}
- Errors: none expected in the normal path

GET /health/live

- Auth: none
- Request: no body, no parameters
- Success: 200 with {"status": "ok"}
- Errors: none expected in the normal path

GET /health/ready

- Auth: none
- Request: no body, no parameters
- Success: 200 with {"status": "ok"}
- Errors: 503 with {"detail": "..."} when readiness checks fail

Auth

GET /v1/auth/nonce

- Auth: none
- Request: query parameter address with a 42-character wallet address
- Success: 200 with {"nonce": "..."}
 - Errors: 422 if the address is malformed

Example response:

```
{
  "nonce": "f1d0b88e3f8b4dce"
}
```

POST /v1/auth/verify

- Auth: none
- Request: body with message and signature
- Success: 200 with access_token, refresh_token, and the authenticated account
- Errors: 401 when the signature or message is invalid

Example request:

```
{
  "message": "127.0.0.1 wants you to sign in with your Ethereum account:\n0x...\n\nURI:
http://127.0.0.1:8000\nVersion: 1\nChain ID: 84532\nNonce: f1d0b88e3f8b4dce\nIssued At:
2026-03-17T12:00:00Z",
  "signature": "0x..."
}
```

Example response:

```
{
  "access_token": "eyJ...",
```

```

    "refresh_token": "eyJ...",
    "account": {
      "id": 1,
      "wallet_address": "0x...",
      "account_type": "human",
      "is_admin": false,
      "display_name": "Demo Account",
      "created_at": "2026-03-17T12:00:00Z",
      "updated_at": "2026-03-17T12:00:00Z"
    }
  }
}

```

POST /v1/auth/refresh

- Auth: none
- Request: body with refresh_token
- Success: 200 with a fresh access_token
- Errors: 401 when the refresh token is invalid or expired

POST /v1/auth/api-keys

- Auth: JWT-only
- Request: body with optional name and expires_at
- Success: 201 with metadata plus the plaintext api_key once
- Errors: 422 when the request is invalid

GET /v1/auth/api-keys

- Auth: JWT-only
- Request: no body
- Success: 200 with a list of API-key metadata
- Errors: none expected in the normal path

DELETE /v1/auth/api-keys/{api_key_id}

- Auth: JWT-only
- Request: path parameter api_key_id
- Success: 204 with no body
- Errors: 404 when the key does not exist

Account

GET /v1/account/me

- Auth: JWT-only
- Request: no body
- Success: 200 with the current account
- Errors: 404 when the account cannot be found

PATCH /v1/account/me

- Auth: JWT-only
- Request: body with optional `display_name`
- Success: 200 with the updated account
- Errors: 422 when no updatable field is provided or the data is invalid

Example request:

```
{
  "display_name": "Demo Provider"
}
```

POST `/v1/account/wallet`

- Auth: JWT-only
- Request: body with `wallet_address`
- Success: 200 with a wallet-change nonce and expiry
- Errors: 409 when the requested change conflicts with account state, 422 for invalid input

POST `/v1/account/wallet/confirm`

- Auth: JWT-only
- Request: body with message and signature
- Success: 200 with fresh tokens and the updated account
- Errors: 409 when the wallet change flow is not valid or has expired

Provider Authoring

POST `/v1/provider/services`

- Auth: generic bearer
- Request: body with `slug`, `name`, `summary`, and optional `description`
- Success: 201 with the created service
- Errors: 409 on conflicts, 422 for invalid input

GET `/v1/provider/services`

- Auth: generic bearer
- Request: no body
- Success: 200 with the authenticated account's services
- Errors: none expected in the normal path

GET `/v1/provider/services/{service_id}`

- Auth: generic bearer
- Request: path parameter `service_id`
- Success: 200 with the owned service detail
- Errors: 404 when the service does not exist or is not owned

PATCH /v1/provider/services/{service_id}

- Auth: generic bearer
- Request: body with optional name, summary, and description
- Success: 200 with the updated service
- Errors: 409 on invalid state transitions, 422 for invalid data

POST /v1/provider/services/{service_id}/tags

- Auth: generic bearer
- Request: body with tags as the full replacement list
- Success: 200 with the service and updated tags
- Errors: 404 if the service is not owned, 422 for invalid input

POST /v1/provider/services/{service_id}/publish

- Auth: generic bearer
- Request: no body
- Success: 200 with the published service
- Errors: 409 when publish requirements are not met

POST /v1/provider/services/{service_id}/endpoints

- Auth: generic bearer
- Request: body with endpoint key, name, summary, description, access_mode, request_schema, response_schema, timeout_seconds, is_enabled, and optional pricing
- Success: 201 with the created endpoint
- Errors: 409 on invalid service state, 422 for schema or pricing problems

PATCH /v1/provider/endpoints/{endpoint_id}

- Auth: generic bearer
- Request: body with optional endpoint fields and optional pricing
- Success: 200 with the updated endpoint
- Errors: 404 if the endpoint is not owned, 409 on invalid state transitions, 422 for invalid input

PUT /v1/provider/endpoints/{endpoint_id}/upstream

- Auth: generic bearer
- Request: body with base_url, path, http_method, and optional config
- Success: 204 with no body
- Errors: 404 if the endpoint is not owned, 409 on invalid state transitions, 422 for invalid input

Example upstream config payload:

```
{
```

```

    "base_url": "https://provider.example.com",
    "path": "/invoke",
    "http_method": "POST",
    "config": {
      "auth": {
        "type": "hmac_sha256",
        "key_id": "gateway-key",
        "secret": "gateway-secret"
      }
    }
  }
}

```

Discovery

GET /v1/services

- Auth: none
- Request: no body
- Success: 200 with the public service list
- Errors: none expected in the normal path

Example response:

```

[
  {
    "id": 1,
    "slug": "demo-agent-service",
    "name": "Demo Agent Service",
    "summary": "Free and paid demo endpoints for manual marketplace testing.",
    "description": "A seeded demo service for discovery, quoting, free invoke, and paid invoke flows.",
    "tags": ["demo", "translation"]
  }
]

```

GET /v1/services/{service_id_or_slug}

- Auth: none
- Request: path parameter service_id_or_slug
- Success: 200 with the public service detail
- Errors: 404 when the service does not exist or is not publicly available
- Lookup rule: a path segment made only of digits is treated as a service id, not a slug

Example response:

```

{
  "id": 1,
  "slug": "demo-agent-service",
  "name": "Demo Agent Service",
  "summary": "Free and paid demo endpoints for manual marketplace testing.",
  "description": "A seeded demo service for discovery, quoting, free invoke, and paid invoke flows.",
  "tags": ["demo", "translation"],
  "endpoints": [
    {
      "key": "free-ping",
      "name": "Free Ping",
      "summary": "A lightweight free endpoint for local demo traffic.",
      "description": "Echoes a simple pong-style response from the mock upstream.",
      "access_mode": "free"
    }
  ]
}

```

```

    },
    {
      "key": "paid-summary",
      "name": "Paid Summary",
      "summary": "A paid endpoint that returns a compact summary.",
      "description": "Uses x402 payment enforcement for paid invocations.",
      "access_mode": "paid"
    }
  ]
}

```

GET /v1/services/{service_id_or_slug}/schema

- Auth: none
- Request: path parameter service_id_or_slug
- Success: 200 with the enabled endpoint schemas
- Errors: 404 when the service does not exist or is not publicly available

Example response:

```

{
  "id": 1,
  "slug": "demo-agent-service",
  "endpoints": [
    {
      "key": "free-ping",
      "request_schema": {
        "type": "object",
        "properties": {
          "message": {"type": "string"}
        },
        "required": ["message"],
        "additionalProperties": false
      },
      "response_schema": {
        "type": "object",
        "properties": {
          "message": {"type": "string"},
          "mode": {"type": "string"}
        },
        "required": ["message", "mode"],
        "additionalProperties": false
      }
    }
  ]
}

```

GET /v1/services/{service_id_or_slug}/pricing

- Auth: none
- Request: path parameter service_id_or_slug
- Success: 200 with the enabled endpoint pricing data
- Errors: 404 when the service does not exist or is not publicly available

Example response:

```

{
  "id": 1,
  "slug": "demo-agent-service",
  "endpoints": [
    {
      "key": "free-ping",

```

```

    "access_mode": "free",
    "pricing_type": "free",
    "amount_minor": null,
    "currency": null
  },
  {
    "key": "paid-summary",
    "access_mode": "paid",
    "pricing_type": "fixed_per_call",
    "amount_minor": 25,
    "currency": "USD"
  }
]
}

```

POST /v1/services/{service_id_or_slug}/quote

- Auth: none
- Request: body with endpoint_key and payload
- Success: 201 with a quote bound to the request payload
- Errors: 404 when the service or endpoint cannot be found, 409 when the quote cannot be created for the current state

Example request:

```

{
  "endpoint_key": "paid-summary",
  "payload": {
    "message": "Summarize this marketplace request."
  }
}

```

Example response:

```

{
  "id": 1,
  "service_id": 1,
  "endpoint_key": "paid-summary",
  "pricing_type": "fixed_per_call",
  "amount_minor": 25,
  "currency": "USD",
  "request_hash": "0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef",
  "service_revision_id": 1,
  "service_change_token":
"0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef",
  "expires_at": "2026-03-17T12:05:00Z",
  "created_at": "2026-03-17T12:00:00Z"
}

```

Invoke

POST /v1/invoke/{service_id_or_slug}

- Auth: generic bearer
- Request: body with endpoint_key, payload, and optional quote_id
- Required header: Idempotency-Key
- Success: 200 with the invocation record, or 402 for unpaid paid-endpoint requests
- Errors: 404 for missing services or invocations, 409 for conflicts or in-progress requests, 502 for upstream failures, 504 for upstream timeouts

Example free-invoke request:

```
{
  "endpoint_key": "free-ping",
  "payload": {
    "message": "hello from the free route"
  }
}
```

Example invocation response:

```
{
  "id": 1,
  "service_id": 1,
  "endpoint_key": "free-ping",
  "access_mode": "free",
  "quote_id": null,
  "idempotency_key": "consumer-example-1",
  "request_hash": "0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef",
  "status": "succeeded",
  "upstream_status_code": 200,
  "response_payload": {
    "message": "pong",
    "mode": "free"
  },
  "error_message": null,
  "created_at": "2026-03-17T12:00:00Z"
}
```

GET /v1/invocations/{invocation_id}

- Auth: generic bearer
- Request: path parameter `invocation_id`
- Success: 200 with the invocation detail
- Errors: 404 when the invocation is not accessible

GET /v1/invocations

- Auth: generic bearer
- Request: no body
- Success: 200 with the caller's invocation list
- Errors: none expected in the normal path

Finance

GET /v1/provider/earnings

- Auth: generic bearer
- Request: no body
- Success: 200 with earnings totals grouped by currency
- Errors: none expected in the normal path

Example response:

```
{
  "totals": [
    {
```

```

        "currency": "USD",
        "charge_minor": 25,
        "platform_fee_minor": 2,
        "provider_earning_minor": 23,
        "entry_count": 1
    }
]
}

```

GET /v1/provider/ledger

- Auth: generic bearer
- Request: no body
- Success: 200 with ledger entries
- Errors: none expected in the normal path

Example response:

```

{
  "entries": [
    {
      "id": 1,
      "service_id": 1,
      "invocation_id": 1,
      "payment_attempt_id": 1,
      "entry_type": "provider_earning",
      "amount_minor": 23,
      "currency": "USD",
      "created_at": "2026-03-17T12:00:00Z"
    }
  ]
}

```

GET /v1/provider/payouts

- Auth: generic bearer
- Request: optional status query parameter
- Success: 200 with payout summaries and payout records
- Errors: none expected in the normal path

Example response:

```

{
  "summaries": [
    {
      "currency": "USD",
      "total_count": 1,
      "ready_count": 1,
      "pending_count": 0,
      "sent_count": 0,
      "failed_count": 0,
      "total_amount_minor": 23,
      "sent_amount_minor": 0
    }
  ],
  "payouts": [
    {
      "id": 1,
      "service_id": 1,
      "invocation_id": 1,
      "payment_attempt_id": 1,
      "destination_wallet": "0xabc...",
    }
  ]
}

```

```

    "amount_minor": 23,
    "currency": "USD",
    "network": "base-sepolia",
    "status": "ready",
    "failure_code": null,
    "attempt_count": 0,
    "created_at": "2026-03-17T12:00:00Z",
    "updated_at": "2026-03-17T12:00:00Z"
  }
]
}

```

POST /v1/provider/payouts

- Auth: JWT-only
- Request: no body, but Idempotency-Key is required
- Success: 200 with payout-request results
- Errors: 409 when a duplicate or conflicting request is submitted

Example response:

```

{
  "idempotency_key": "example-provider-payout-1",
  "requested_count": 2,
  "sent_count": 2,
  "failed_count": 0,
  "payouts": []
}

```

Admin

POST /v1/admin/services/{service_id}/suspend

- Auth: admin actor
- Request: body with reason
- Success: 201 with the moderation action
- Errors: 404 when the service cannot be found, 409 for invalid transitions

Example request:

```

{
  "reason": "Provider maintenance window."
}

```

Example response:

```

{
  "id": 1,
  "service_id": 1,
  "actor_account_id": 1,
  "action": "suspend",
  "reason": "Provider maintenance window.",
  "created_at": "2026-03-17T12:00:00Z"
}

```

POST /v1/admin/services/{service_id}/restore

- Auth: admin actor
- Request: body with reason

- Success: 201 with the moderation action
- Errors: 404 when the service cannot be found, 409 for invalid transitions

POST /v1/admin/services/{service_id}/delist

- Auth: admin actor
- Request: body with reason
- Success: 201 with the moderation action
- Errors: 404 when the service cannot be found, 409 for invalid transitions

GET /v1/admin/moderation/actions

- Auth: admin actor
- Request: service_id query parameter
- Success: 200 with the moderation action history
- Errors: 422 for invalid query values

Example response:

```
[
  {
    "id": 1,
    "service_id": 1,
    "actor_account_id": 1,
    "action": "suspend",
    "reason": "Provider maintenance window.",
    "created_at": "2026-03-17T12:00:00Z"
  },
  {
    "id": 2,
    "service_id": 1,
    "actor_account_id": 1,
    "action": "restore",
    "reason": "Maintenance completed.",
    "created_at": "2026-03-17T12:30:00Z"
  }
]
```

Implementation Notes

- Quote creation is public. Authenticated clients may still call it, but

consumers do not need bearer credentials to ask for a quote.

- Discovery routes only expose public, active service data.
- Provider upstream URLs and upstream credentials are never returned by public discovery responses.
- The paid invoke flow is a two-step 402 challenge/settle flow backed by x402-compatible headers.
- The provider payout request path is separate from payout reporting and requires JWT auth plus idempotency protection.